TechKnights — Federated Traffic-Flow Prediction

Experiment Evaluation Report

Configuration

Dataset	METR-LA
Distribution	Non-IID
Clients	3
Rounds completed	20
Seed	42
Speed	■ Normal

Summary Metrics

Final Federated MAE	4.4390
Final Federated RMSE	7.8400
Final Centralized MAE	5.1980
Final Centralized RMSE	8.4640
Convergence round	Round 4
Total communication (federated)	5158.12 KB
Communication per round	257.91 KB/round

Communication Cost Comparison

Federated total (cumulative, actual)	5.04 MB
Centralized total (one-time, 60 sensors × 70% train split)	5.49 MB
Model payload per transfer	42.98 KB
Reduction vs Centralized	8.3%

Round-wise Metrics

Round	Fed MAE	Fed RMSE	Cen MAE	Cen RMSE	Comm (KB)
1	5.5210	9.0930	4.6610	7.9730	257.91
2	4.4620	8.0020	4.4720	7.7250	257.91
3	4.3840	7.9020	4.3750	7.8130	257.91
4	4.5830	7.9740	5.4230	8.8230	257.91
5	4.5900	7.9470	4.8330	8.0010	257.91
6	4.6060	7.9360	4.5140	7.8450	257.91
7	4.5710	8.0280	4.3710	7.7560	257.91
8	4.6270	8.0760	4.4450	7.8800	257.91

9	4.4630	7.8940	4.7340	7.9880	257.91
10	4.4400	7.9700	4.9800	8.3000	257.91
11	4.3770	7.8520	4.8640	8.2310	257.91
12	4.6810	8.2120	4.7650	8.0560	257.91
13	4.7000	8.1110	4.9140	8.2200	257.91
14	4.3780	7.8620	4.4960	8.0010	257.91
15	4.6890	8.0780	4.4660	7.9310	257.91
16	4.4210	7.8290	5.1490	8.3200	257.91
17	4.3940	7.8840	4.4790	7.9130	257.91
18	4.3810	7.8580	4.8020	8.1910	257.91
19	4.6750	7.9680	4.8470	8.2530	257.91
20	4.4390	7.8400	5.1980	8.4640	257.91

Client Performance (final round)

Client	Final MAE
Client 1	4.8500
Client 2	4.9270
Client 3	5.4490

Federated vs Centralized

Metric	Federated	Centralized	Gap
MAE	4.4390	5.1980	-14.6%
RMSE	7.8400	8.4640	-7.4%
Privacy	No raw data shared	Raw data centralised	—
Bandwidth	Weights only	Full raw stream	—